

Lab Goal : This lab was designed to teach you more about linked lists. In this lab a `SinglyLinkedList` object stores the list, unlike last lab where the list was created in the main method and passed in as a parameter to the methods.

Lab Description : Use the `ListNode` class given below to write some basic Linked List methods. In this implementation only a reference to the first node is stored in a reference variable. You access the other blocks by iterating to them from the first node.

ListNode – stores a value and a reference to the next node

```
public class ListNode implements Linkable
{
    private Comparable listNodeValue;
    private ListNode nextListNode;

    public ListNode() {
        listNodeValue = null;
        nextListNode = null;
    }

    public ListNode(Comparable value, ListNode next) {
        listNodeValue = value;
        nextListNode = next;
    }

    public Comparable getValue() {
        return listNodeValue;
    }

    public ListNode getNext() {
        return nextListNode;
    }

    public void setValue(Comparable value) {
        listNodeValue = value;
    }

    public void setNext(Linkable next) {
        nextListNode = (ListNode)next;
    }
}
```

Sample Data :

See the main of lab15c.

Sample Output :

```
Original list values:  go on at 34 2.1 -a-2-1 up over
                      num nodes = 8
List values after calling nodeCount:  go on at 34 2.1 -a-2-1 up over
List values after calling doubleFirst: go go on at 34 2.1 -a-2-1 up over
List values after calling doubleLast:  go go on at 34 2.1 -a-2-1 up over over
List values after calling skipEveryOther: go on 34 -a-2-1 over
List values after calling removeXthNode (2): go 34 over
List values after calling setXthNode (2,one): go one over
List values after calling removeXthNode (1):
```